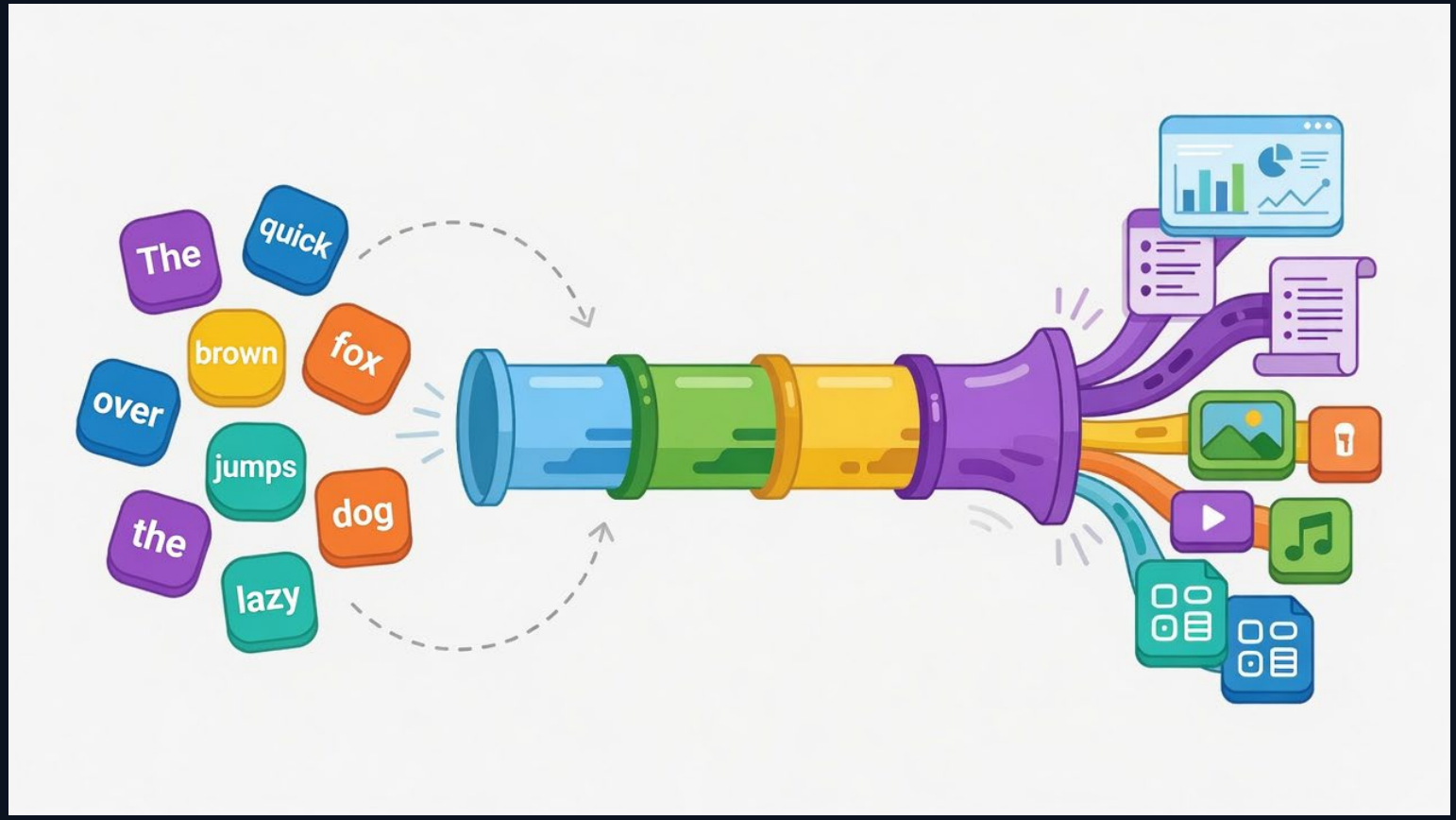
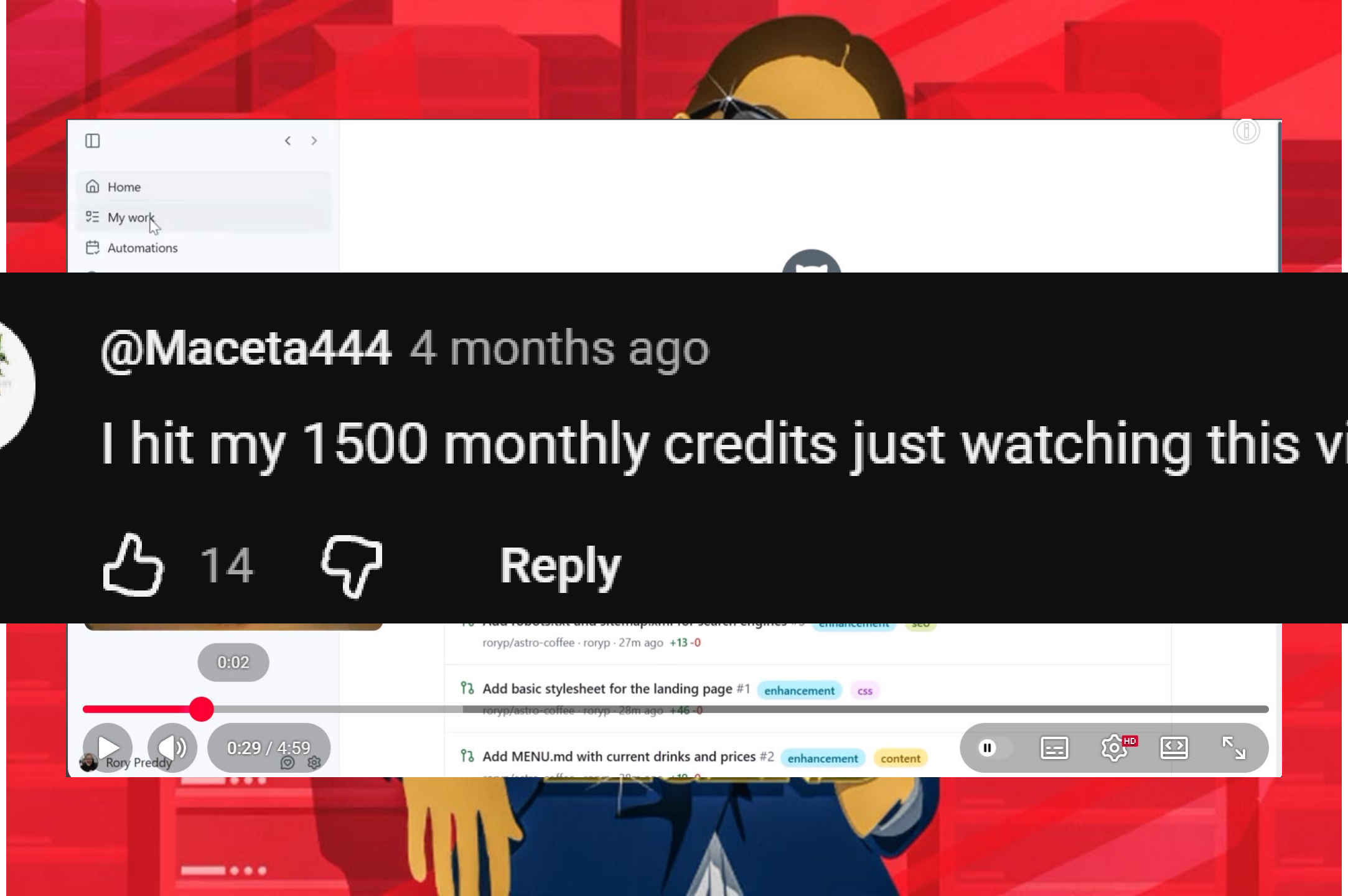
Token Economics



Agenda

- Introduction
- Demo 1 - 4 levers
- Demo 2 - Pricing
- Tracing and Scaling
- Resources


Home
My work
Automations
@Maceta444 4 months ago
I hit my 1500 monthly credits just watching this video
14
Reply
0:02
0:29 / 4:59
Rory Preddy
roryp/astro-coffee · roryp · 27m ago +13 -0
Add basic stylesheet for the landing page #1
enhancement
css
roryp/astro-coffee · roryp · 28m ago +46 -0
Add MENU.md with current drinks and prices #2
enhancement
content
HD

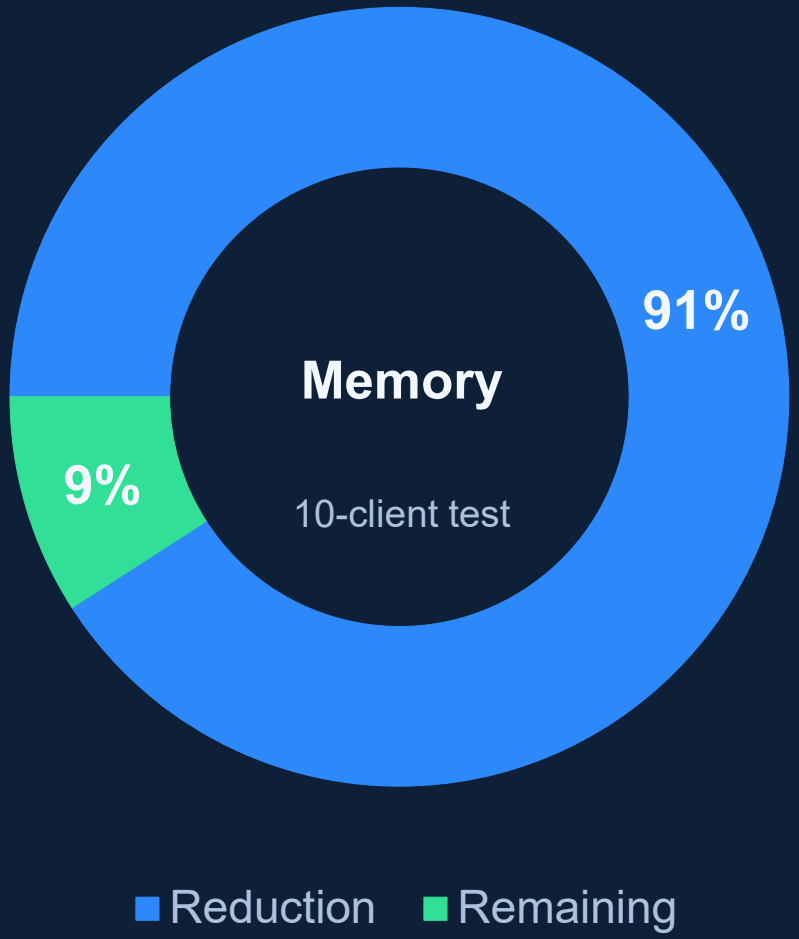
GitHub Copilot Runtime Ported to Rust

A faster, leaner foundation for agents



https://aka.ms/copilot-migration

Copilot runtime performance



91%

less memory

1,383 MB before / 126 MB after

65%

less processor time

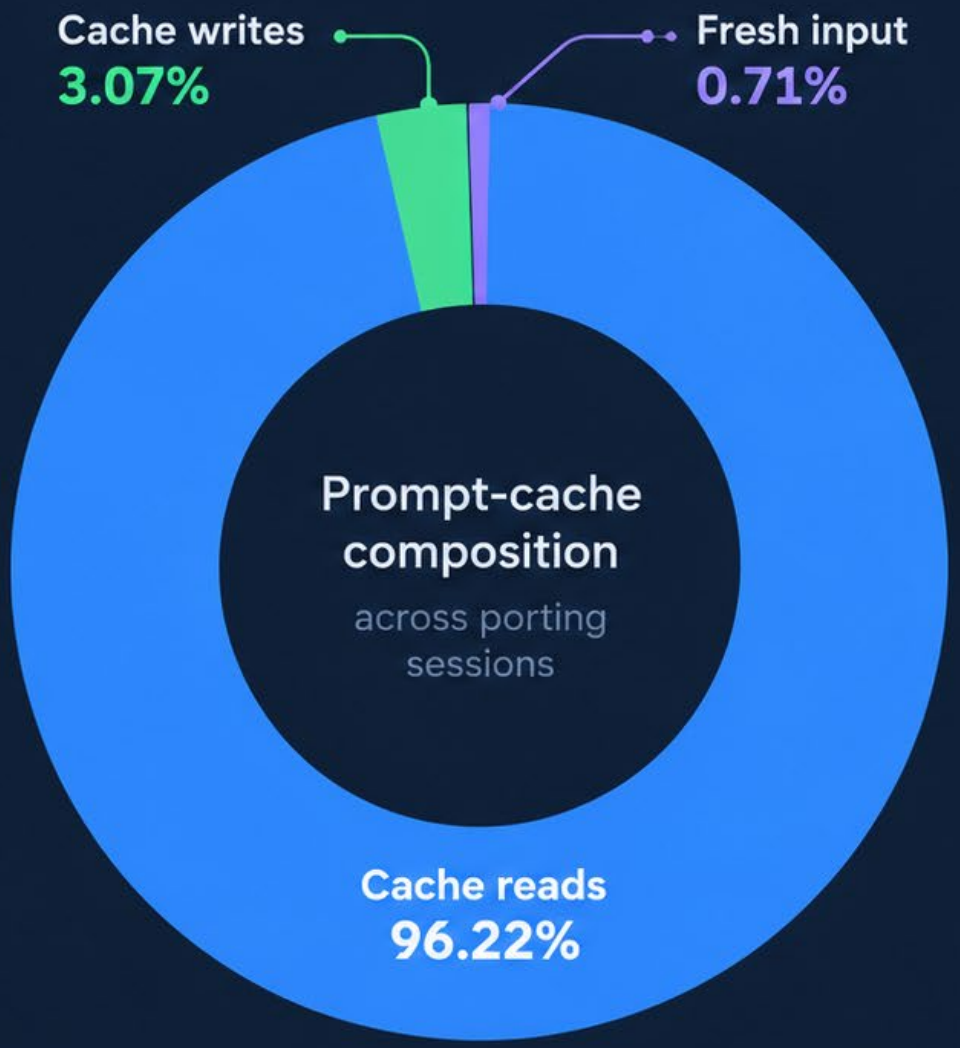
312 s before / ~110 s after

4×

faster session lifecycle

5.25 s before / 1.33 s after

Cost savings



$120K
total spend

$0.14
per line of production Rust

136.3B
total tokens

130.6B cached reads · **4.2B** cache writes · **0.9B** fresh input · **0.6B** output

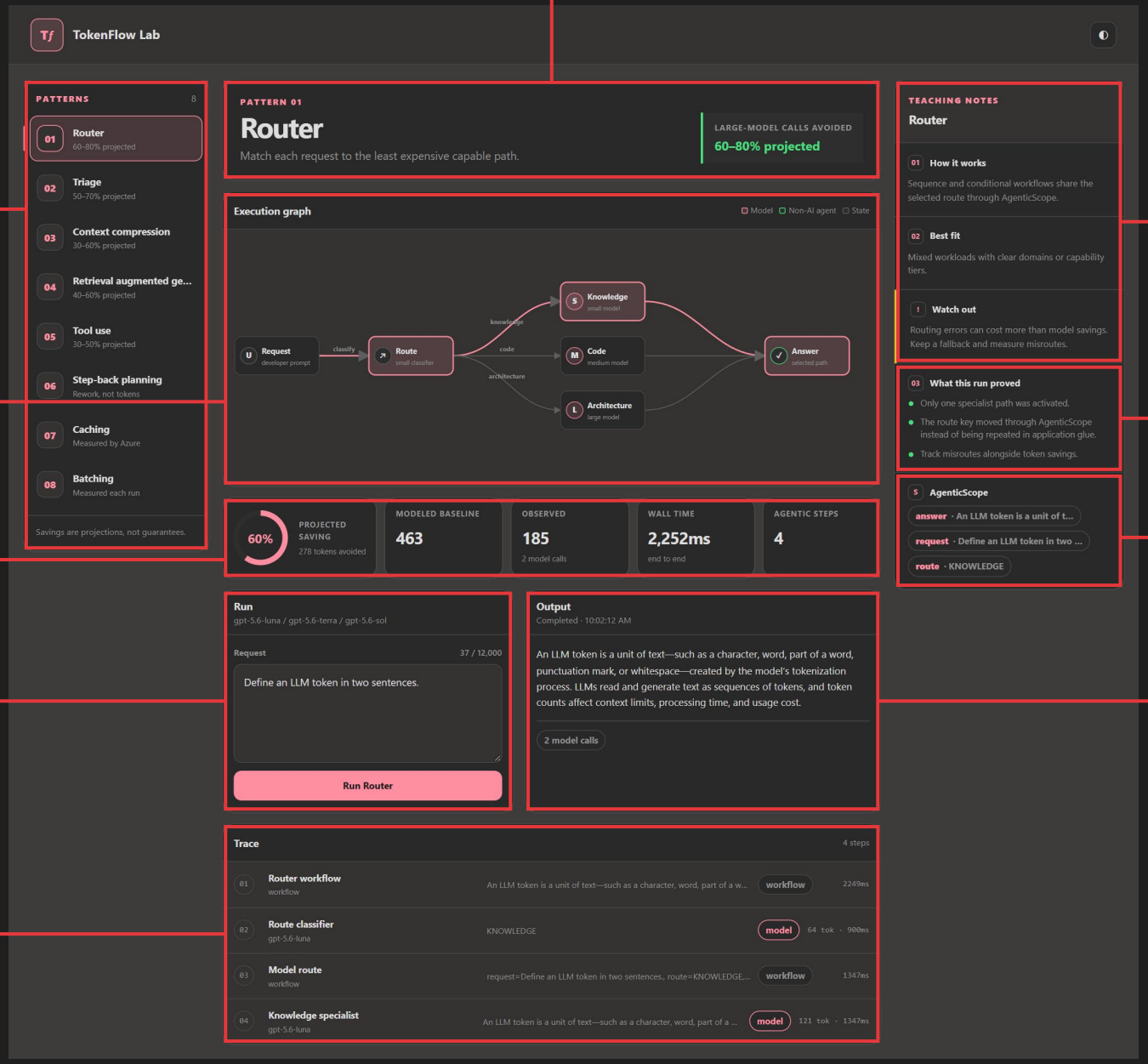
96.22% prompt-cache hit rate — primary savings lever

128 PRs

Agenda

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TOKENFLOW LAB



1. Patterns

2. Pattern + goal

3. Execution graph

4. Token metrics

5. Request + run

6. Answer

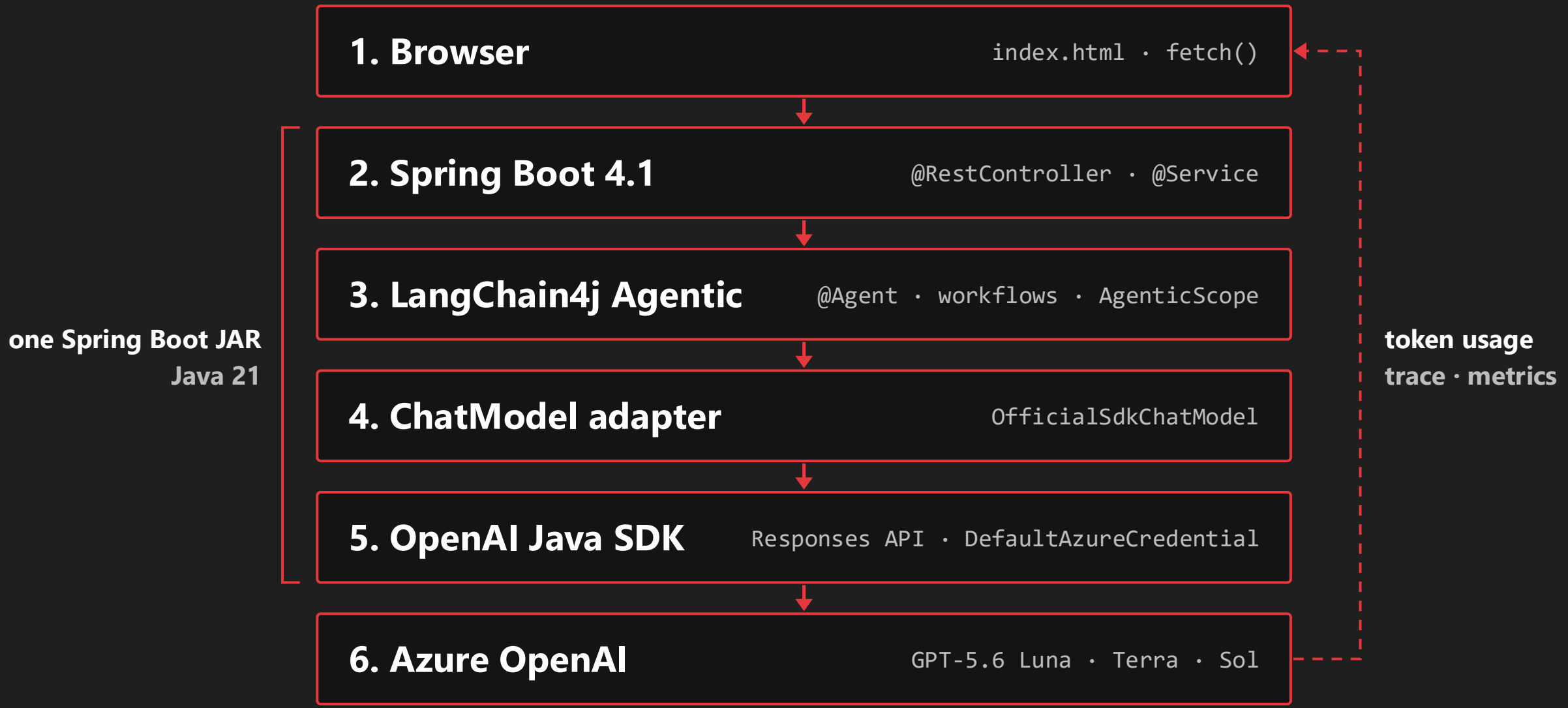
7. Trace

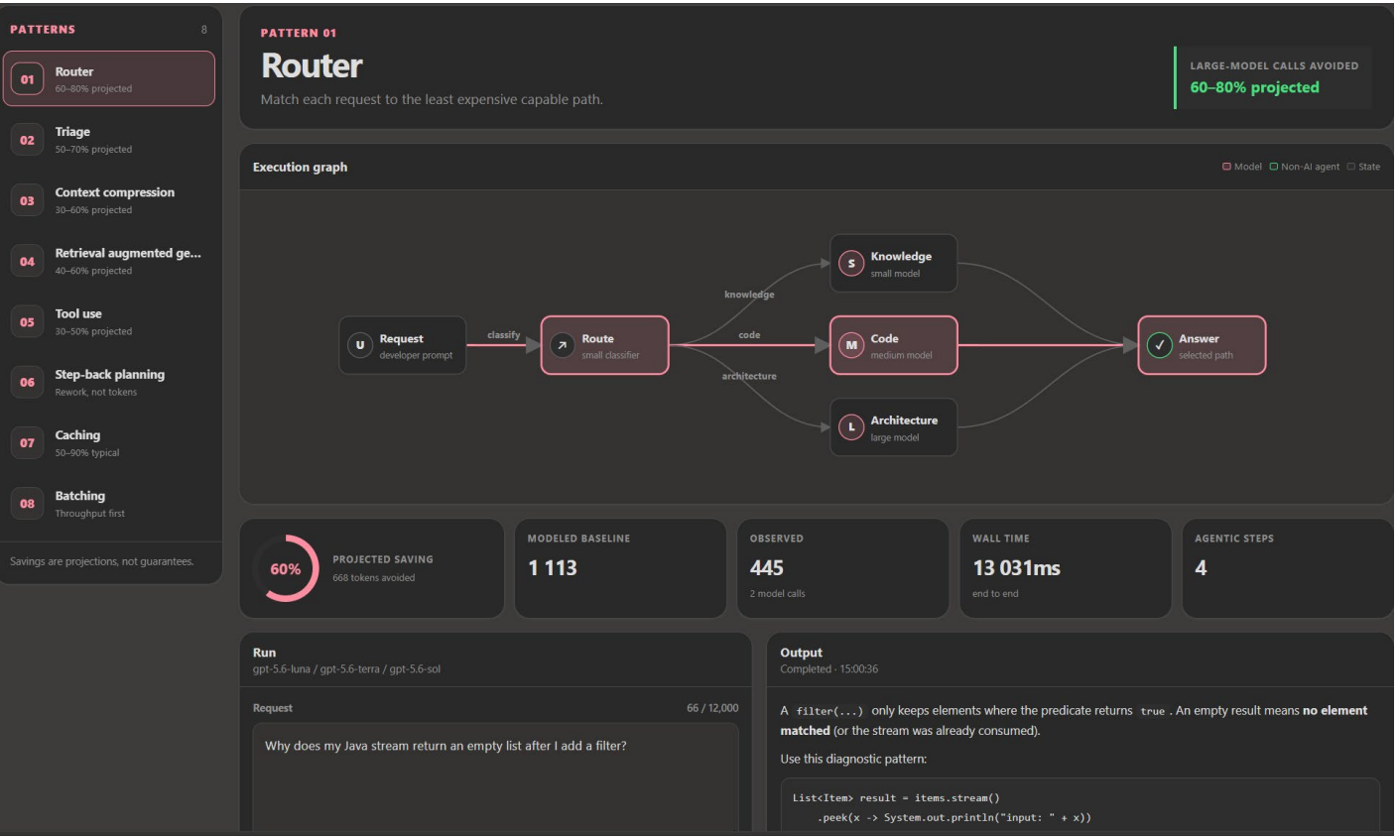
8. Teaching notes

9. What it proved

10. AgenticScope

SPRING + LANGCHAIN4J

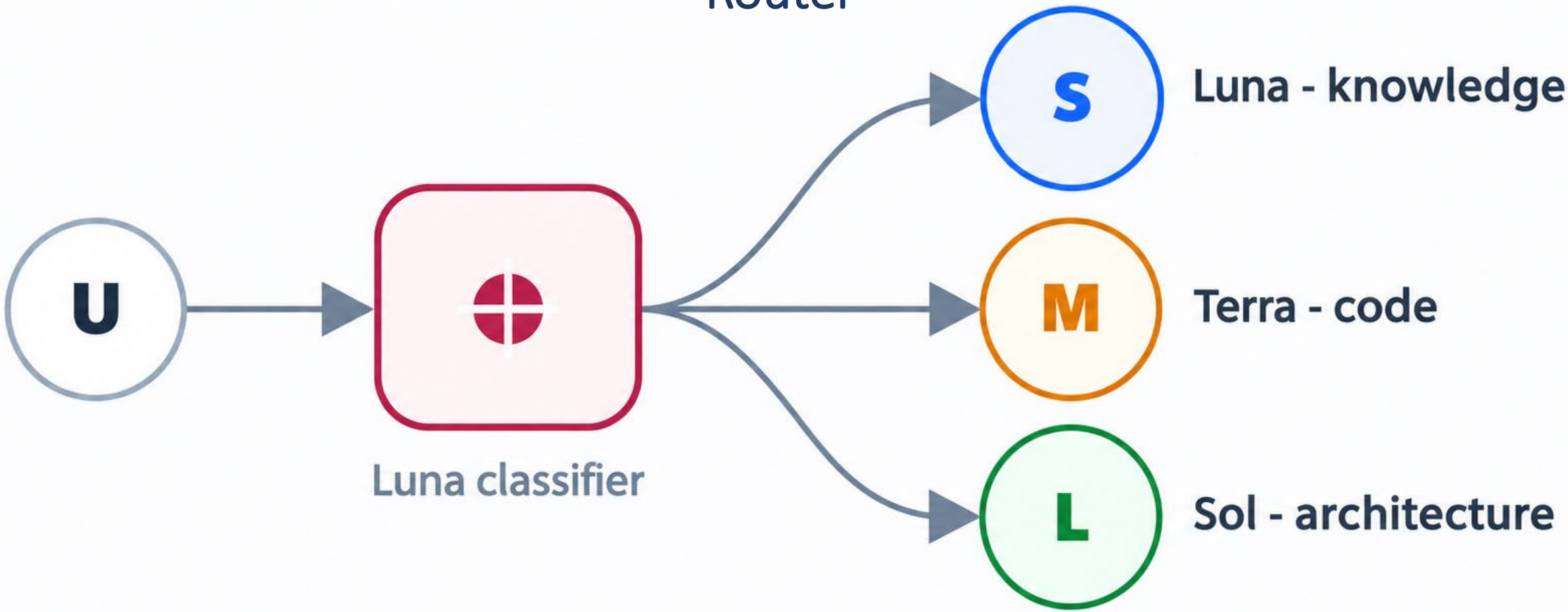




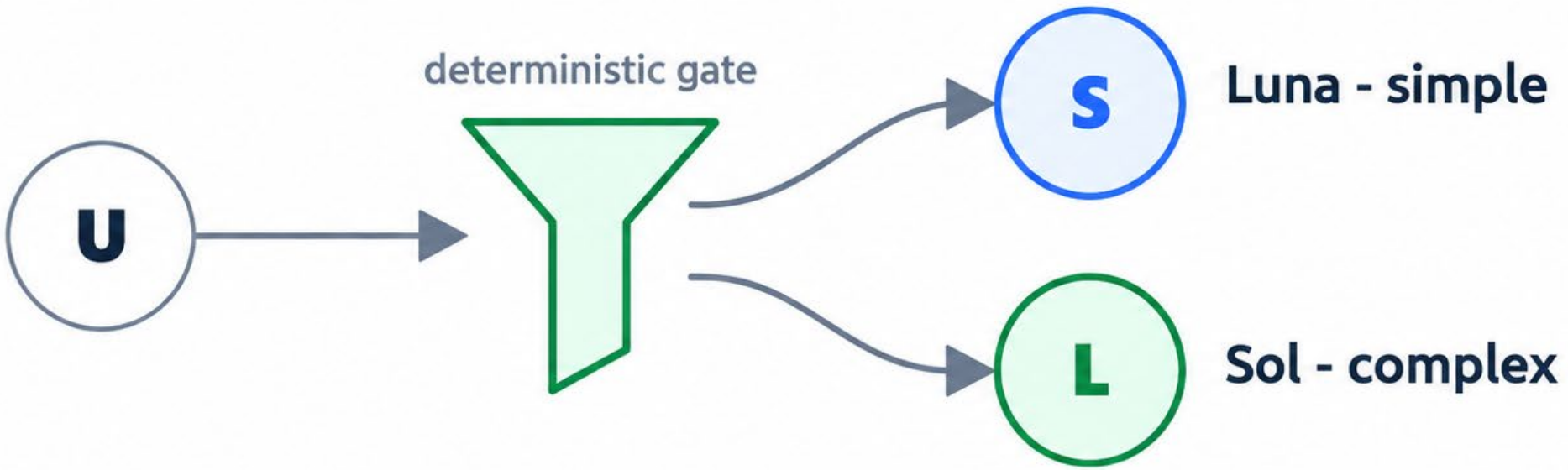
https://aka.ms/costs


MODEL
Router
Triage
PROMPT
Step-back planning
CONTEXT
Compression
RAG
Tool use
Caching
OUTPUT
Batching

Router

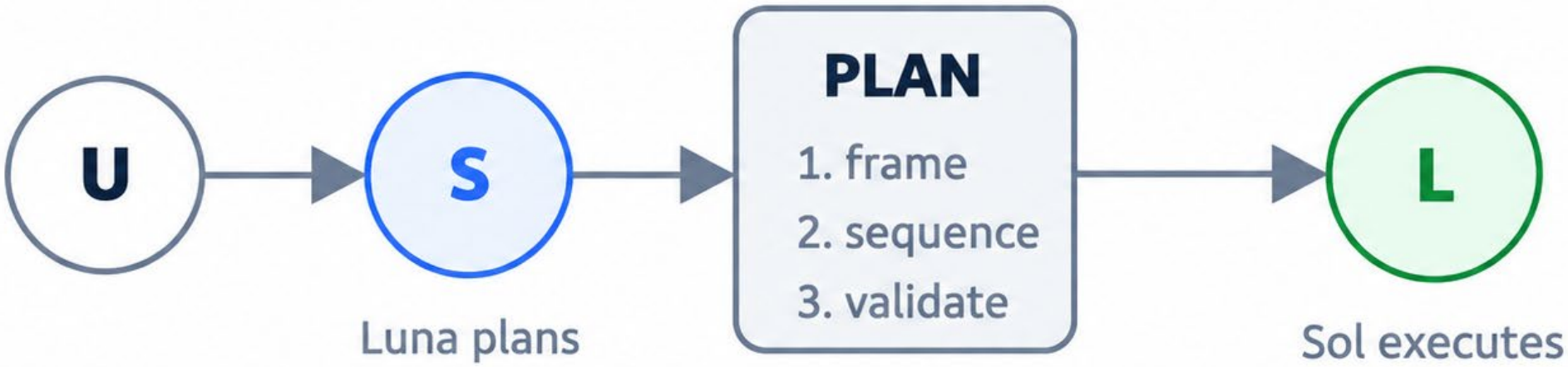

U
Luna classifier
S
Luna - knowledge
M
Terra - code
L
Sol - architecture

Triage



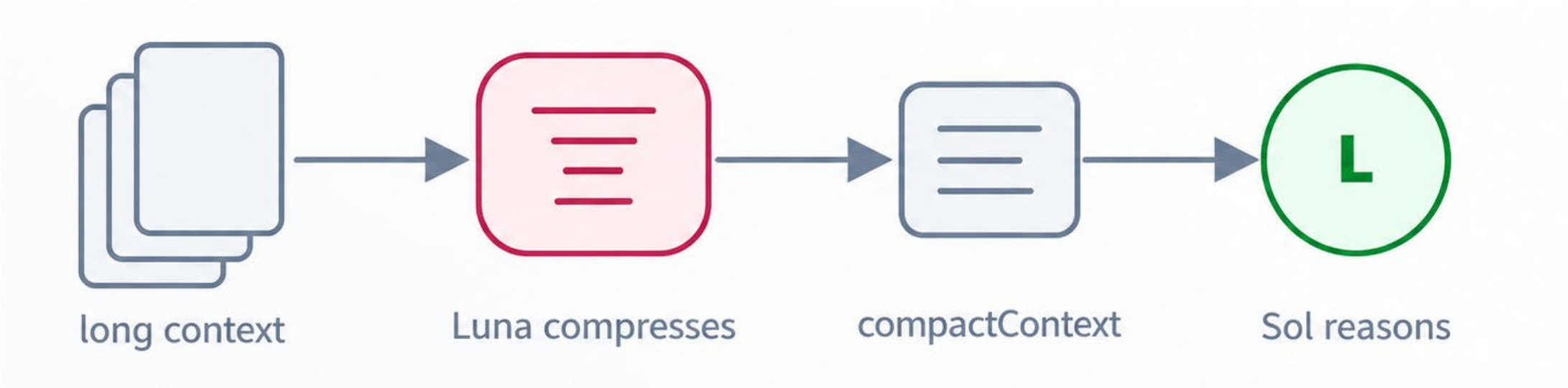

MODEL
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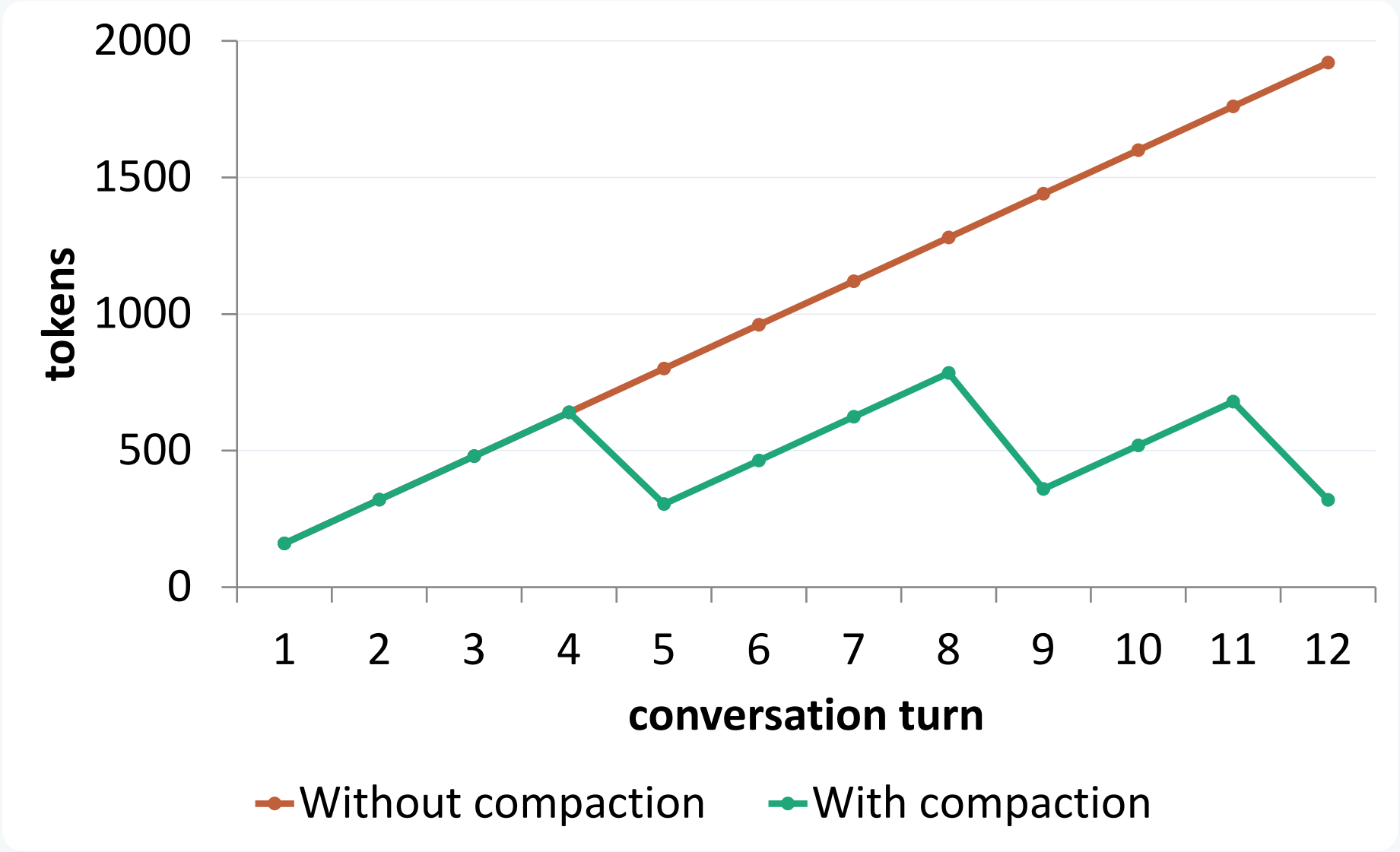
Step-back planning



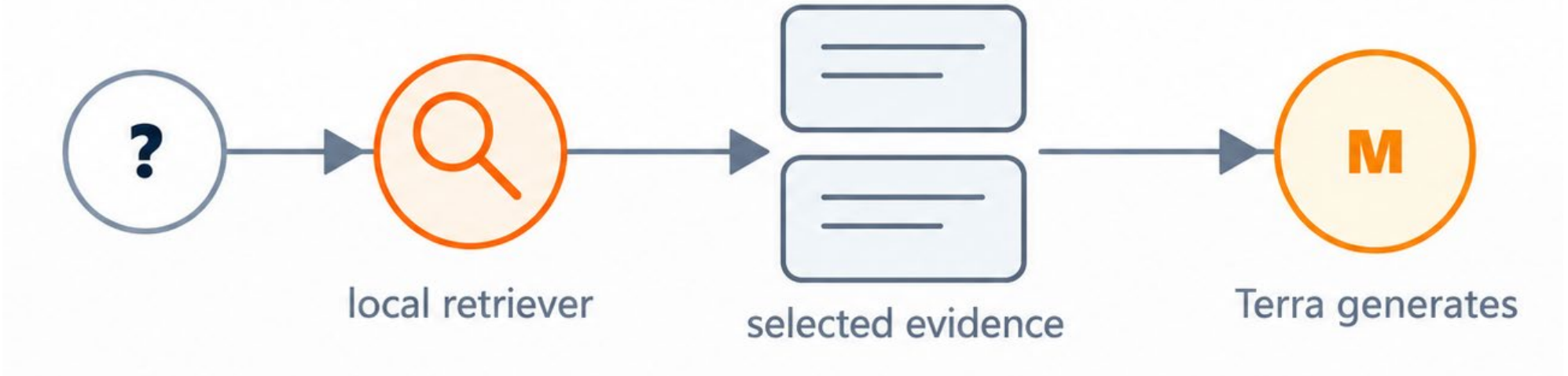

MODEL
PROMPT
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Compression

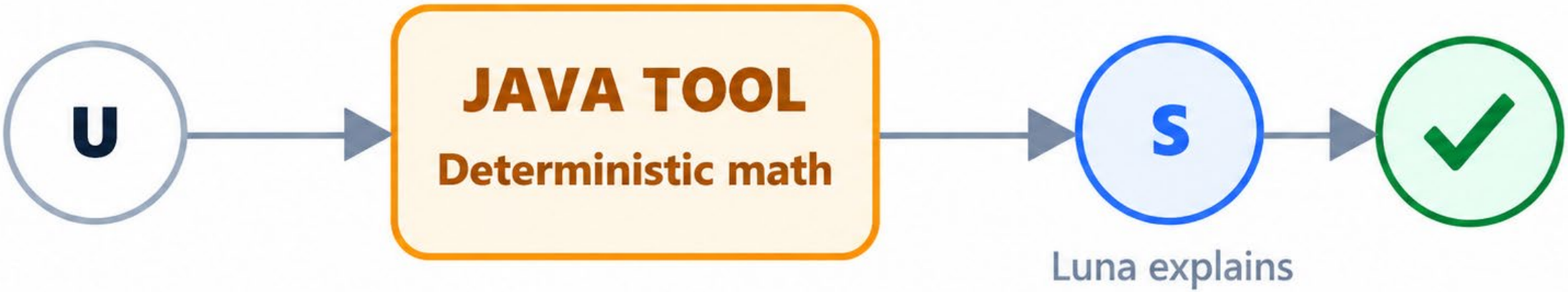



2000
1500
1000
500
0
tokens
1
2
3
4
5
6
7
8
9
10
11
12
conversation turn
Without compaction
With compaction

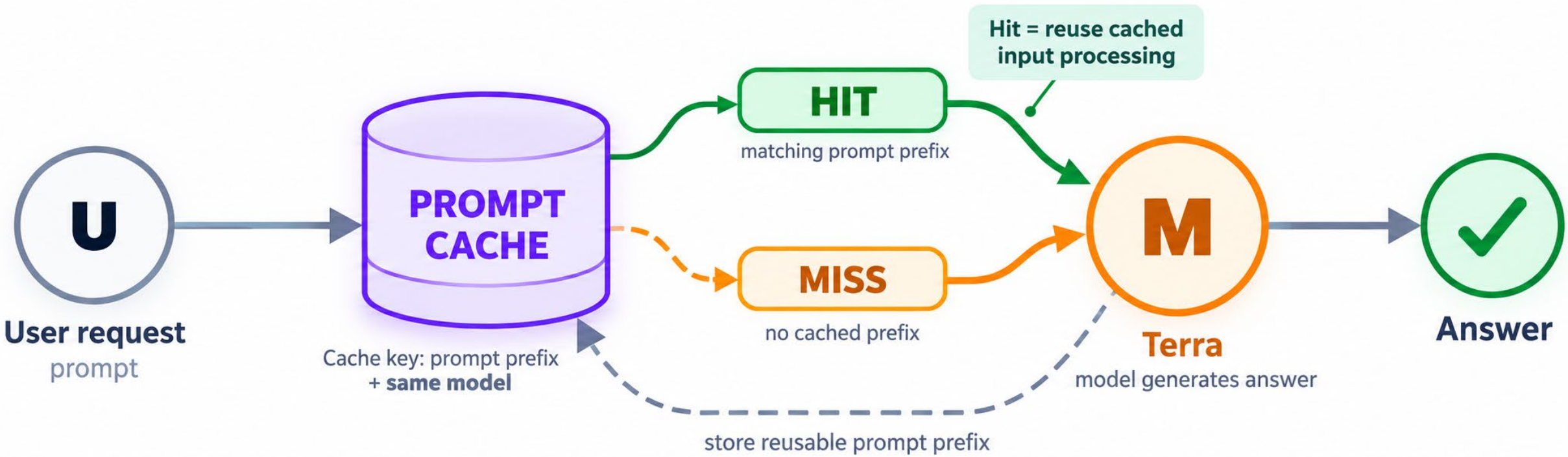
Retrieval-Augmented Generation (RAG)



Tools

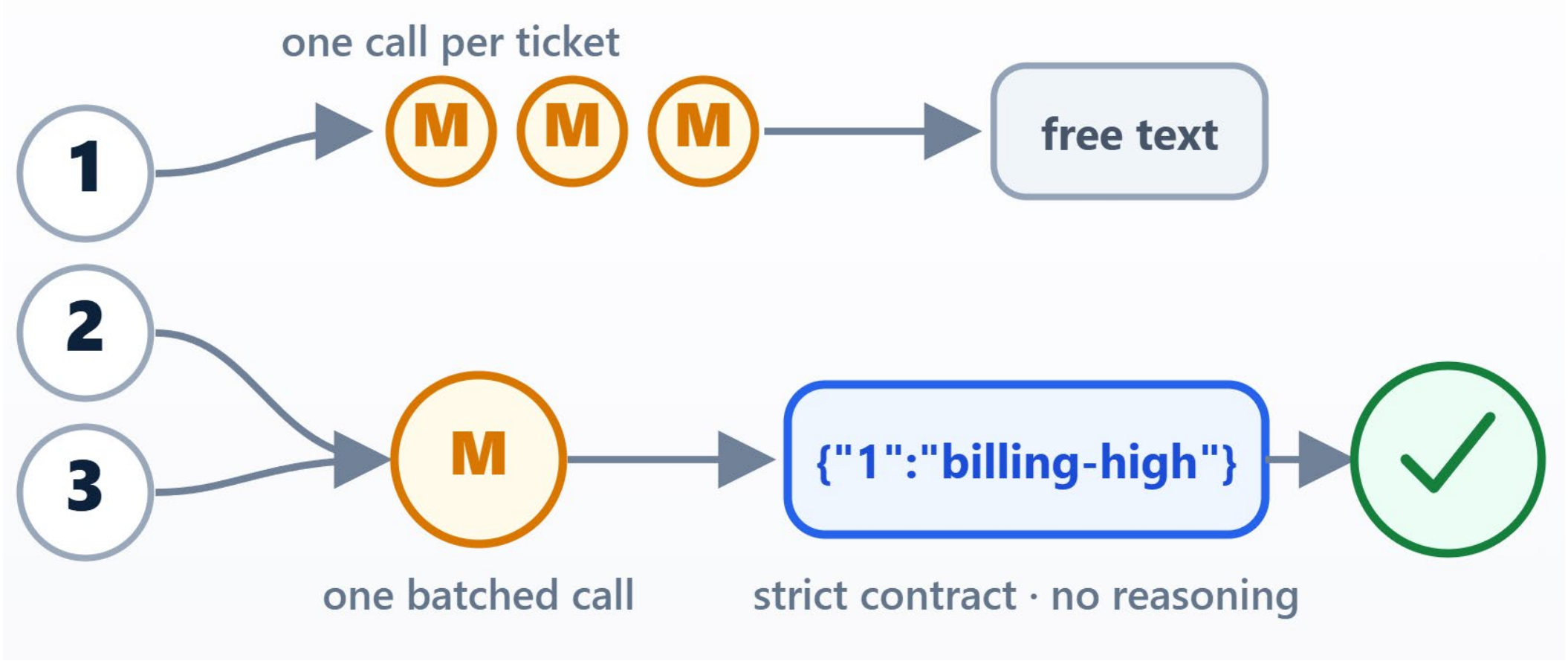


Caching


Hit = reuse cached input processing
HIT
matching prompt prefix
PROMPT CACHE
U
User request
prompt
MISS
no cached prefix
M
Terra
model generates answer
Answer
Cache key: prompt prefix + same model
store reusable prompt prefix


MODEL
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Tool use
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Batching

Batching


one call per ticket
1
M
M
M
free text
2
3
M
{"1":"billing-high"}
one batched call
strict contract · no reasoning


MODEL
PROMPT
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OUTPUT
Router
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= 4 tokens

Your sentence

Seven shapes make words into falling blocks.

8 tokens / o200k_base 44/500


Seven


␣shapes


␣make


␣words


␣into


␣falling


␣blocks


.

Join game


{
"BillingCurrency": "USD",
"CustomerEntityId": "Default",
"CustomerEntityType": "Retail",
"Items": [
{
"currencyCode": "USD",
"tierMinimumUnits": 0,
"retailPrice": 1.2,
"unitPrice": 1.2,
"armRegionName": "eastus",
"location": "US East",
"effectiveStartDate": "2026-08-01T00:00:00Z",
"meterId": "c6713056-7cef-508c-9e2e-7035984a4f44",
"meterName": "5.6 luna ShortCo Opt Std Gl 1M Tokens",
"productId": "DZH318Z0T9WD",
"skuId": "DZH318Z0T9WD/1166",
"productName": "Azure OpenAI GPT5",
"skuName": "5.6 luna ShortCo Opt Std Gl",
"serviceName": "Foundry Models",
"serviceId": "DZH31600VF1F",
"serviceFamily": "AI + Machine Learning",
"unitOfMeasure": "1M",
"type": "Consumption",
"isPrimaryMeterRegion": true,
"armSkuName": "56 luna ShortCo Opt Std Gl"
}
],
"NextPageLink": null,
"Count": 1
}
Currency: billing currency.
Retail price: list price charged.
Unit price: price per billing unit.
Region code: Azure region identifier.
Region name: human-friendly region name.
Effective date: when this price became active.
Meter name: full billing meter description.
Product name: product family.
SKU name: specific model/tier variant.
Service name: Azure service category.
Unit of measure: billed per 1M tokens.
Type: consumption-based billing.

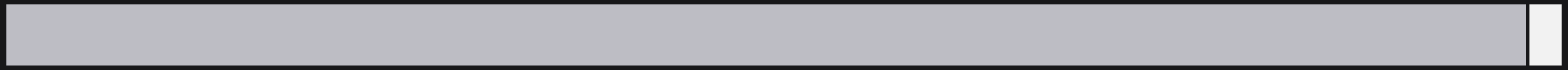
COST OF ONE MOVE: LUNA, TERRA AND SOL

The same plain move, with no cache: about 10,000 input and 35 output tokens at each model's prices

	INPUT TOKENS	PER 1M INPUT	OUTPUT TOKENS	PER 1M OUTPUT	PRICES ARE PER 1M TOKENS	PER MOVE
Luna gpt-5.6-luna	(10,000 ×	$0.20 +	35 ×	$1.20) ÷	1,000,000 ≈	**$0.002**
Terra gpt-5.6-terra	(10,000 ×	$2.00 +	35 ×	$12.00) ÷	1,000,000 ≈	**$0.020**
Sol gpt-5.6-sol	(10,000 ×	$4.00 +	35 ×	$20.00) ÷	1,000,000 ≈	**$0.041**

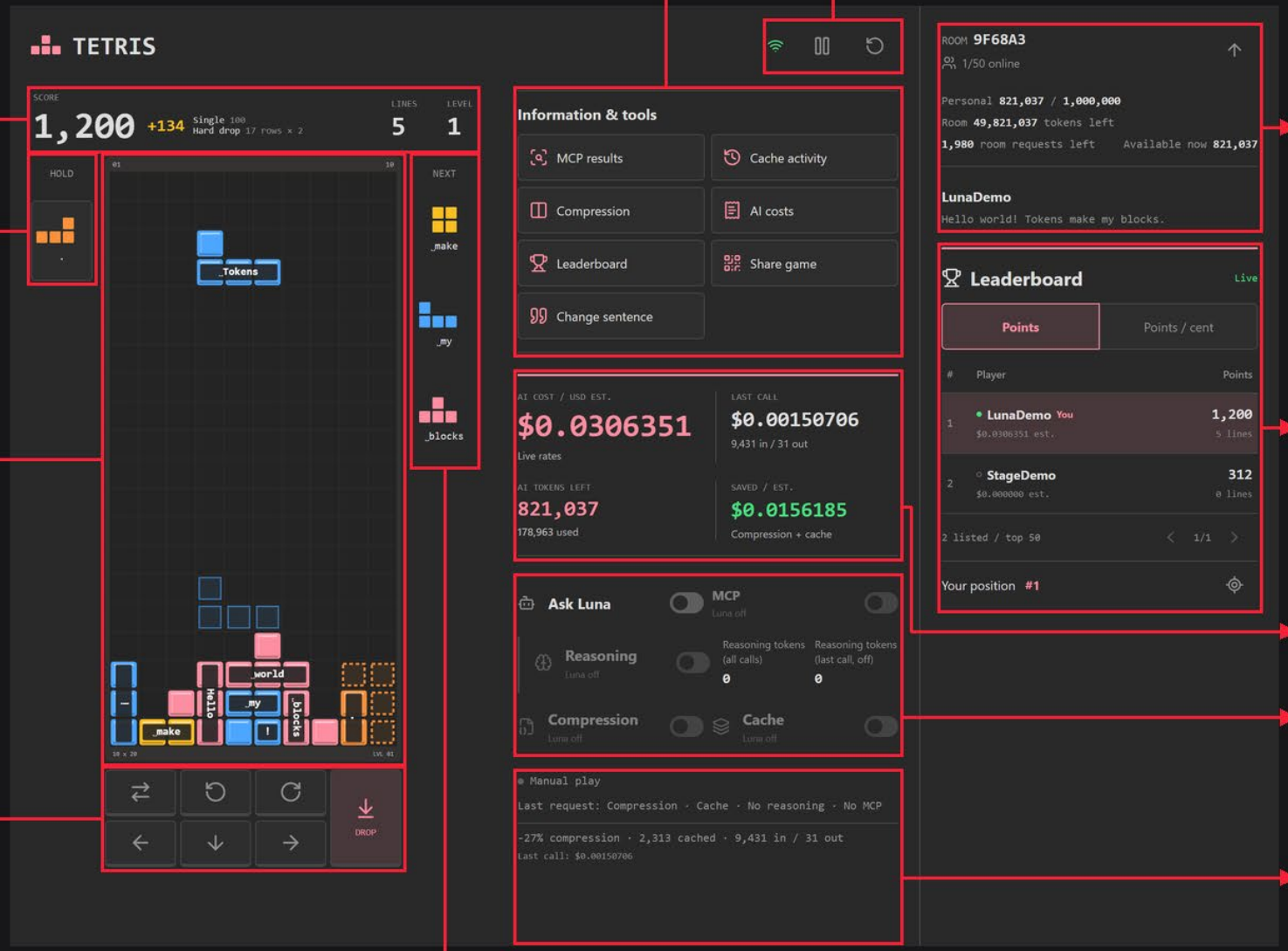
WHERE THE COST GOES

INPUT $0.002000 OUTPUT $0.000042



Input is about 98% of that

TETRIS + LUNA



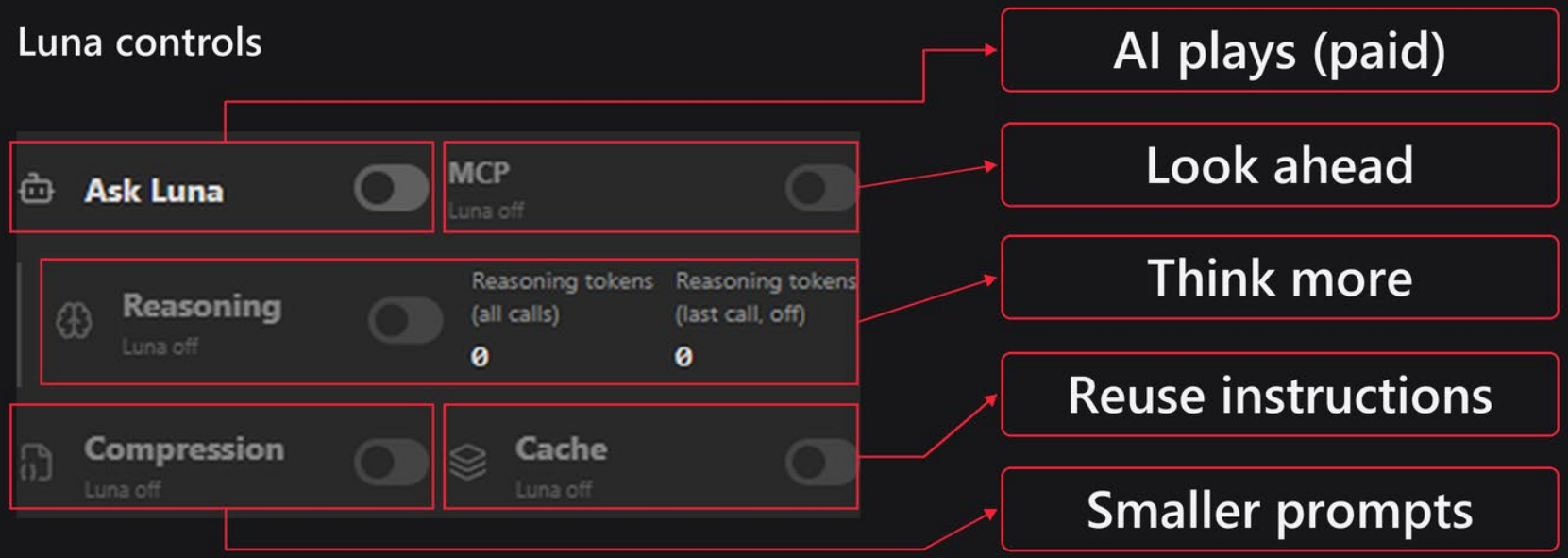
1. Pause / Restart
2. Score + points
3. Hold
4. Board
5. Next
6. Controls
7. Tools
8. AI cost
9. AI play
10. Status
11. Room
12. Rankings

AI COSTS + LUNA CONTROLS

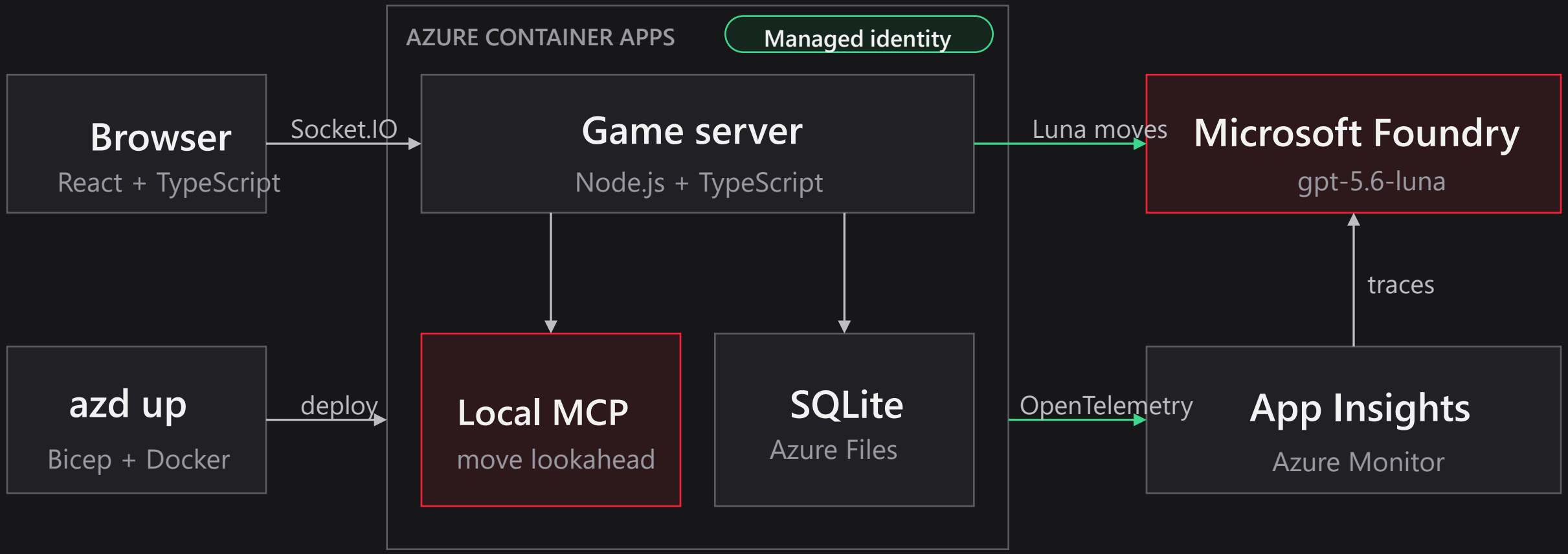
AI costs (USD estimates)

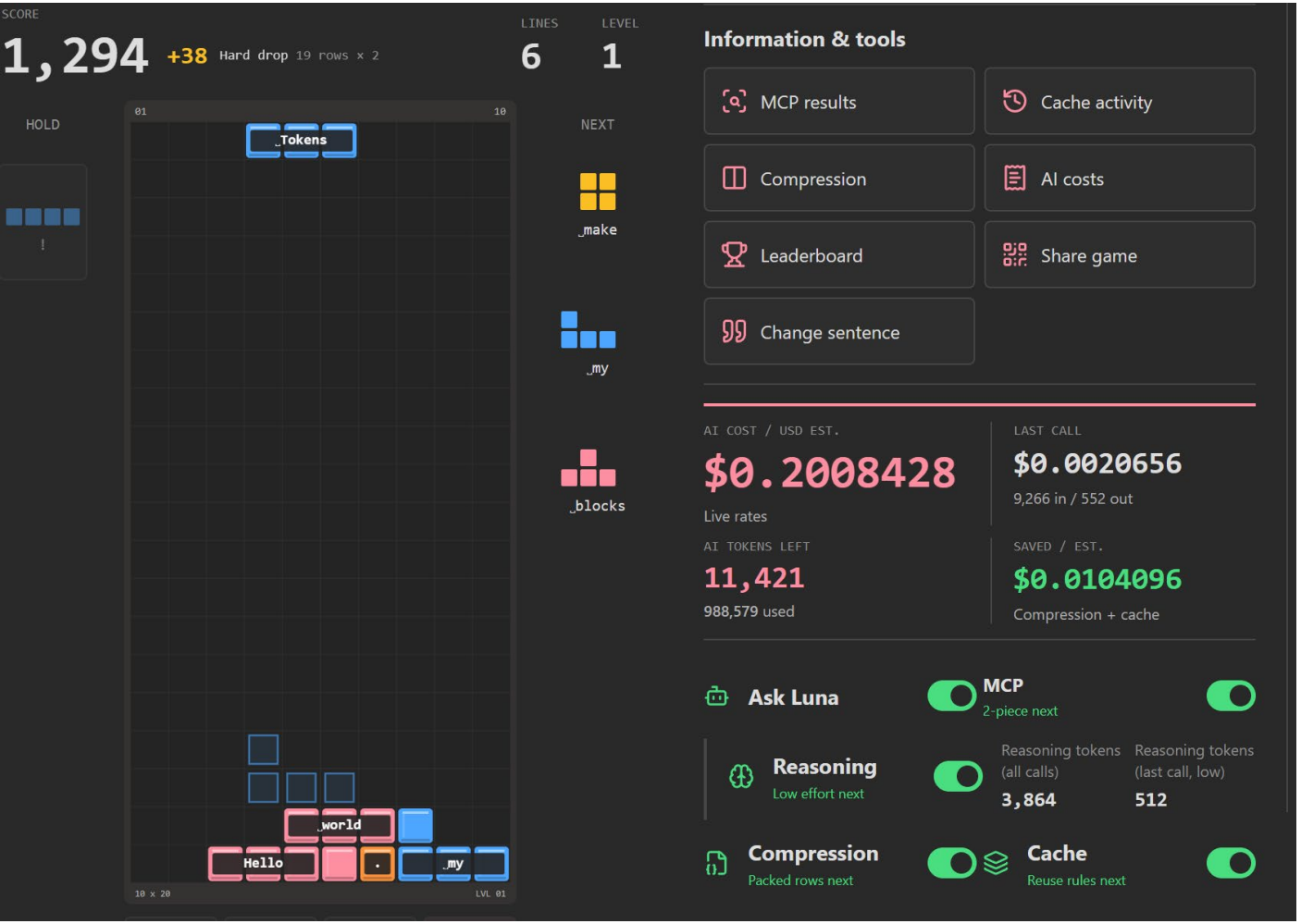


Luna controls



Architecture



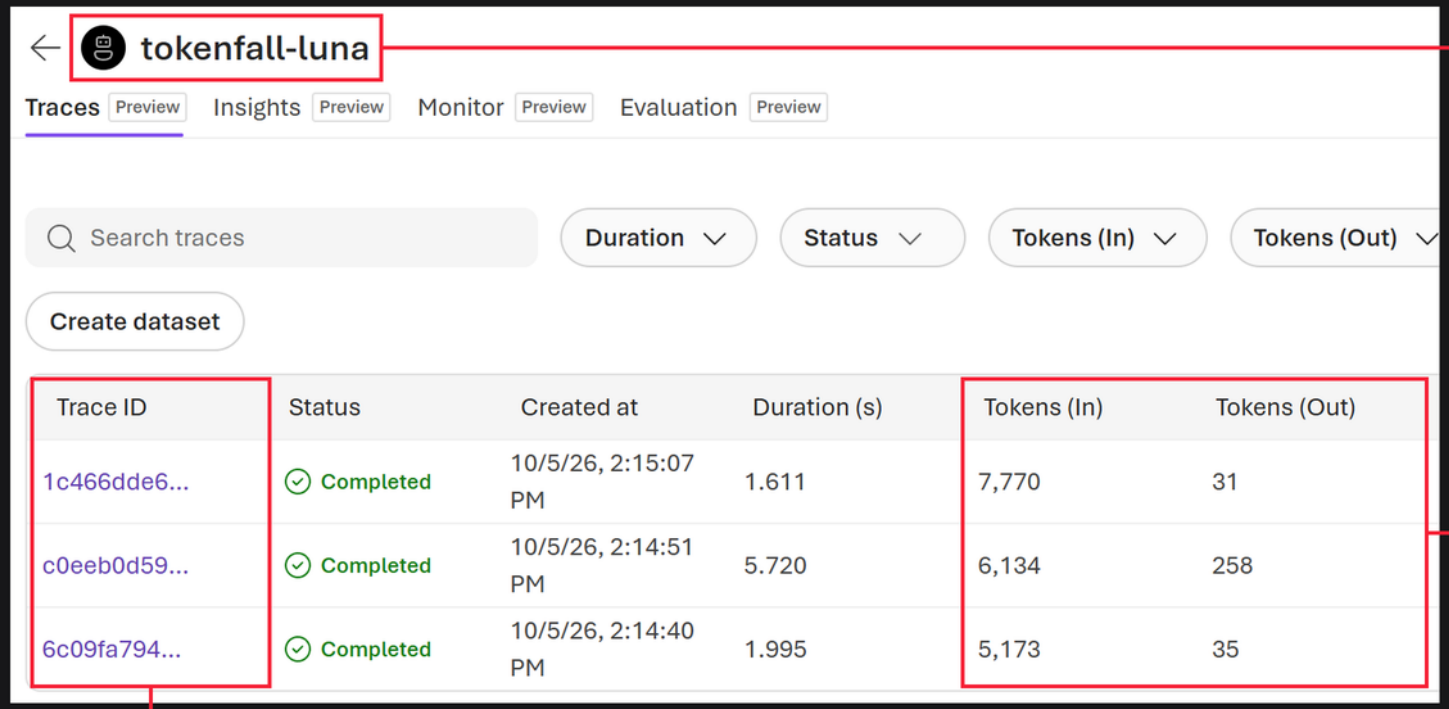


https://aka.ms/tokenfall

Agenda

- Introduction
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Tracing

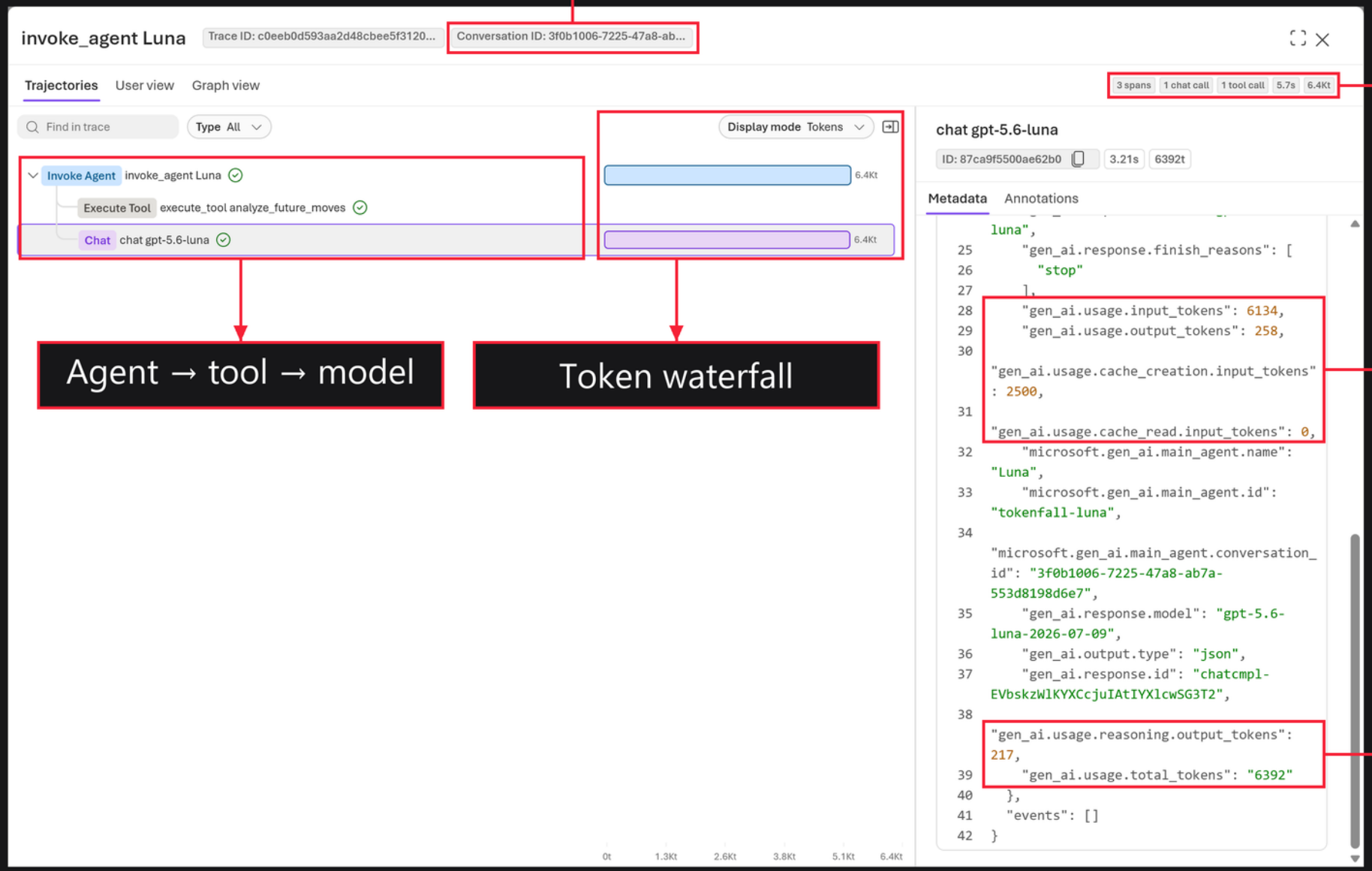


Linked Luna agent

Tokens in / out

One trace per move

One Luna move: MCP lookahead + reasoning



Agent → tool → model

Token waterfall

Game run

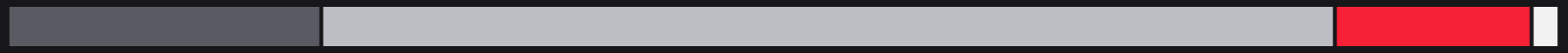
Trace totals

Input, output, cache

Reasoning + total

WHERE MCP'S $2.25 GOES

Cost of 1,000 Luna moves with MCP, averaged from 10 Foundry traces



	TOKENS PER MOVE	× PRICE PER 1M	= PER 1,000 MOVES	
Instructions Game rules and reply format	2,276	$0.20	**$0.45**	Same every move: cacheable
Game state Board, pieces and every legal move	7,352	$0.20	**$1.47**	Most of the cost: 65%
MCP lookahead Tool results for the next two moves	1,427	$0.20	**$0.29**	What MCP adds: +15%
Luna's reply The chosen move and a short tip	34	$1.20	**$0.04**	Output: 6× price per token
Total	11,089		**$2.25**	= $0.00225 per move

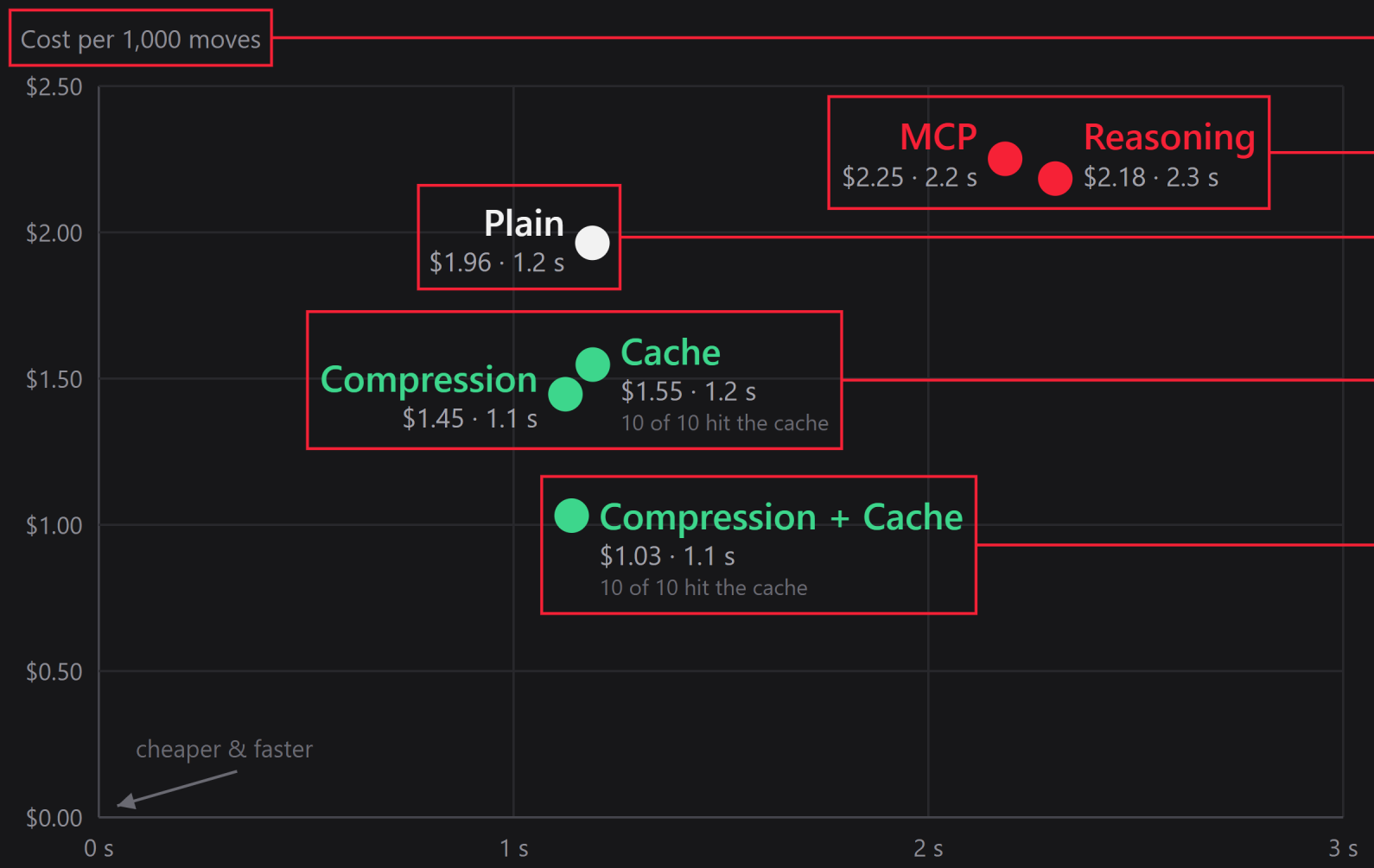
TIME PER MOVE



Tool: +0.6 s, no AI call
was 2.1 s on 0.5 CPU

LUNA COST VS SPEED

Same 10 board positions for every option · game server with 2 CPUs



1 move ≈ $0.001–$0.002

Costs more, 1.8–1.9× slower

Baseline: no options

Cheaper, same speed

Together: 47% cheaper

Foundry trace duration

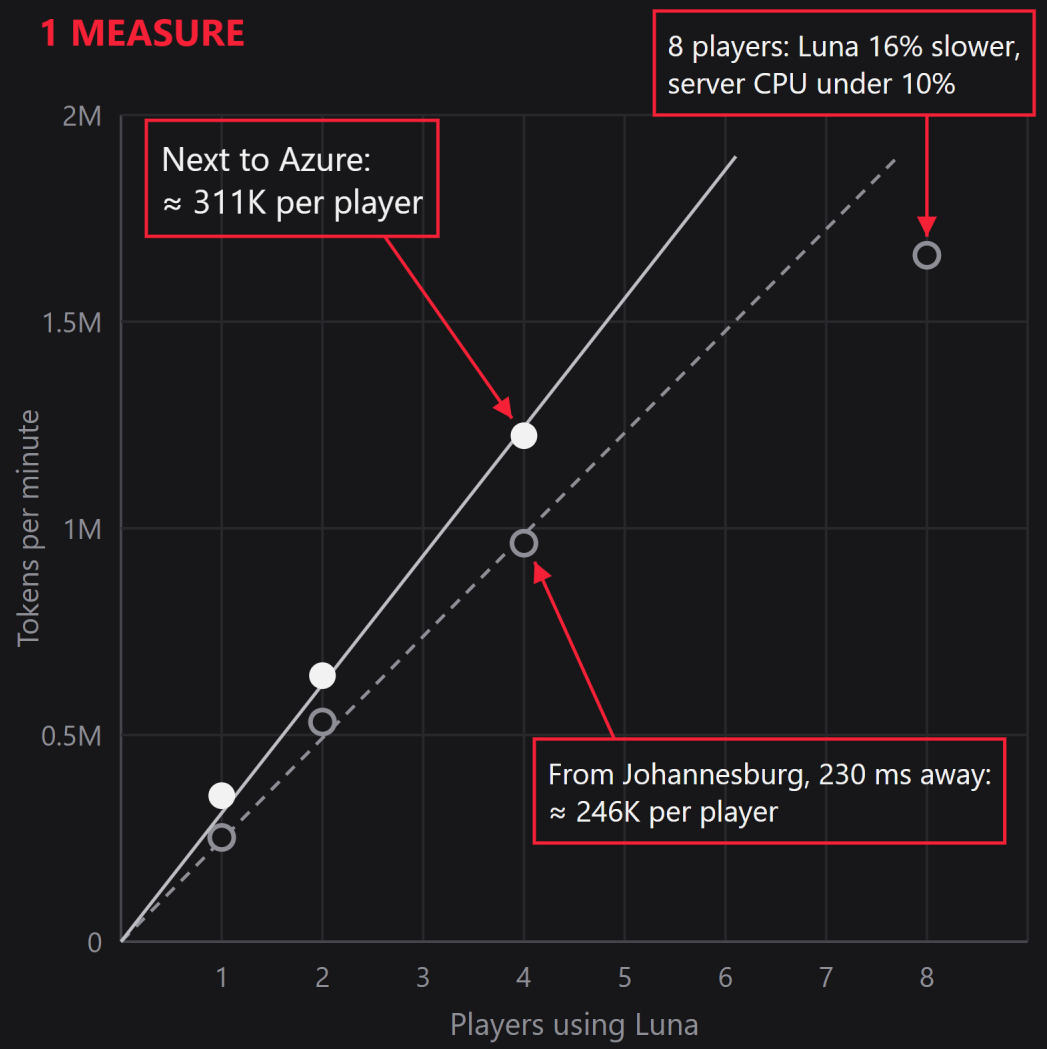
Microsoft Foundry traces · gpt-5.6-luna · Azure retail prices

Tokens × Azure price

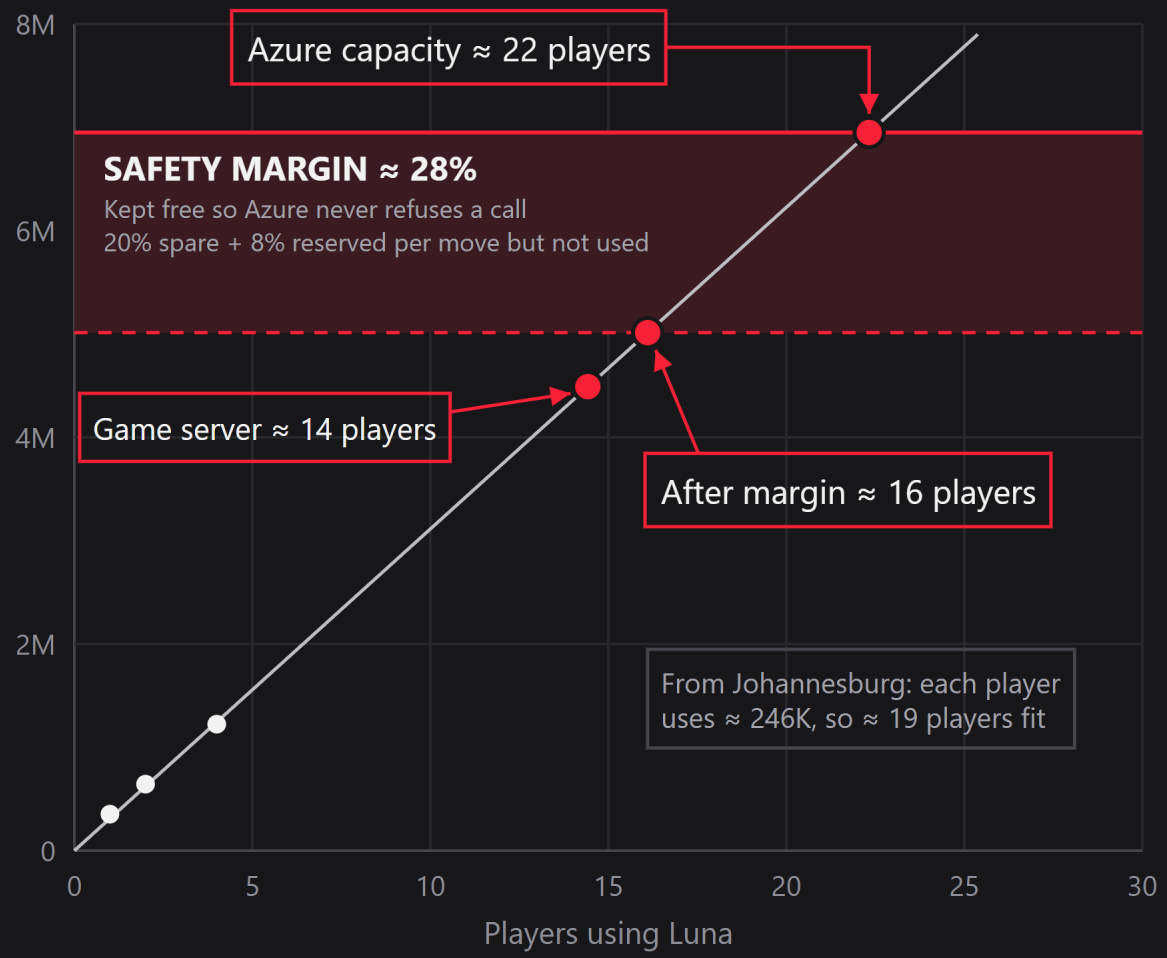
HOW MANY LUNA PLAYERS AT ONCE?

Plain Luna moves on the live game server: 1 replica, 2 CPUs, 4 GiB

1 MEASURE

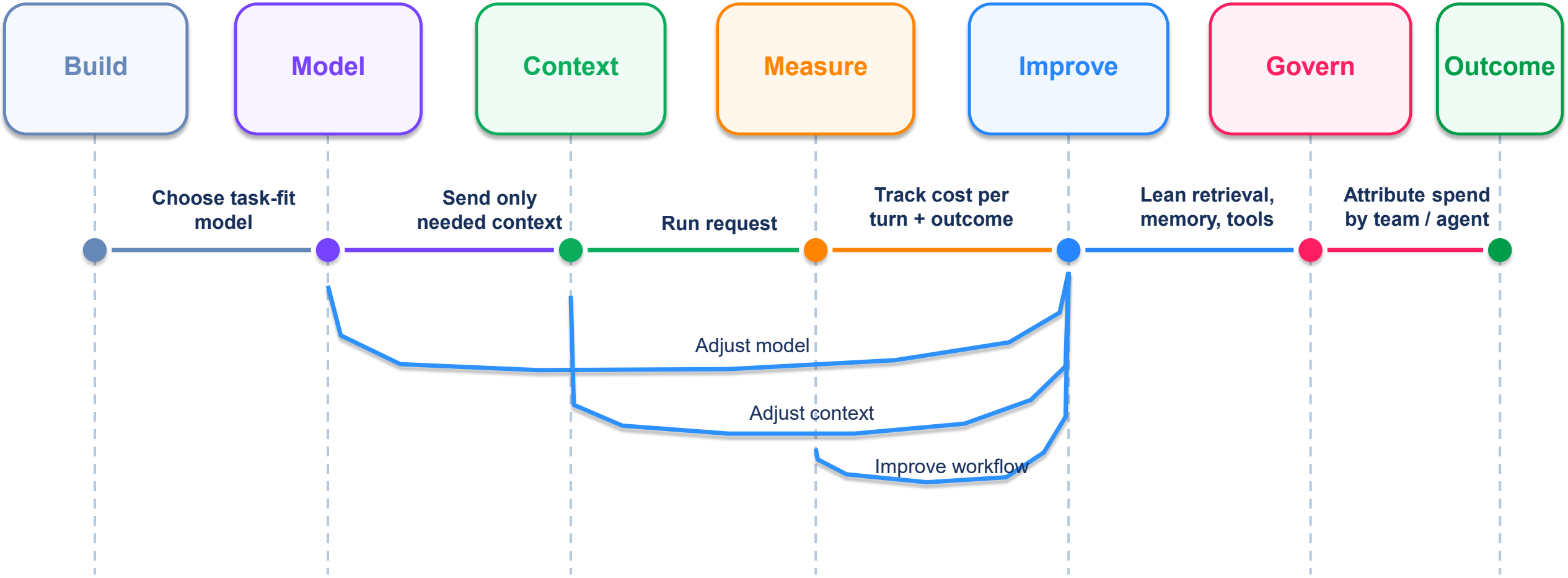


2 PREDICT



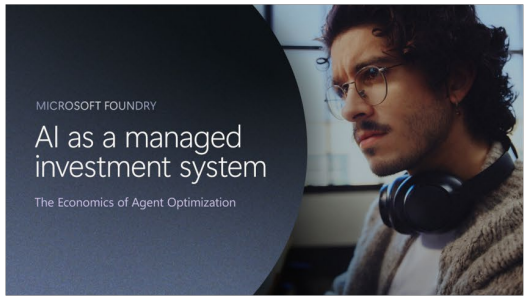
Plain moves, one minute per step · Next to Azure: bots inside the app's container · Johannesburg: bots on the internet, 230 ms away

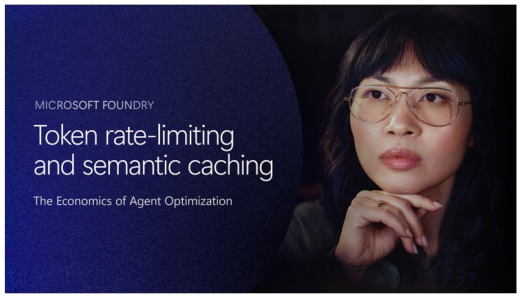
Token Economics Checklist

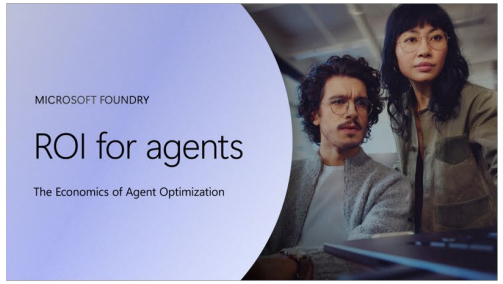


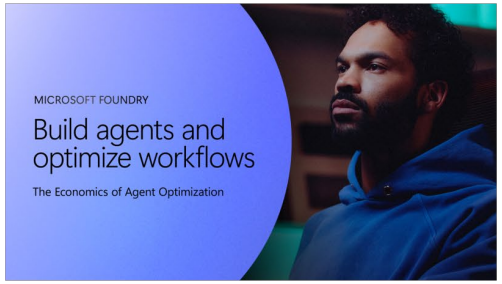
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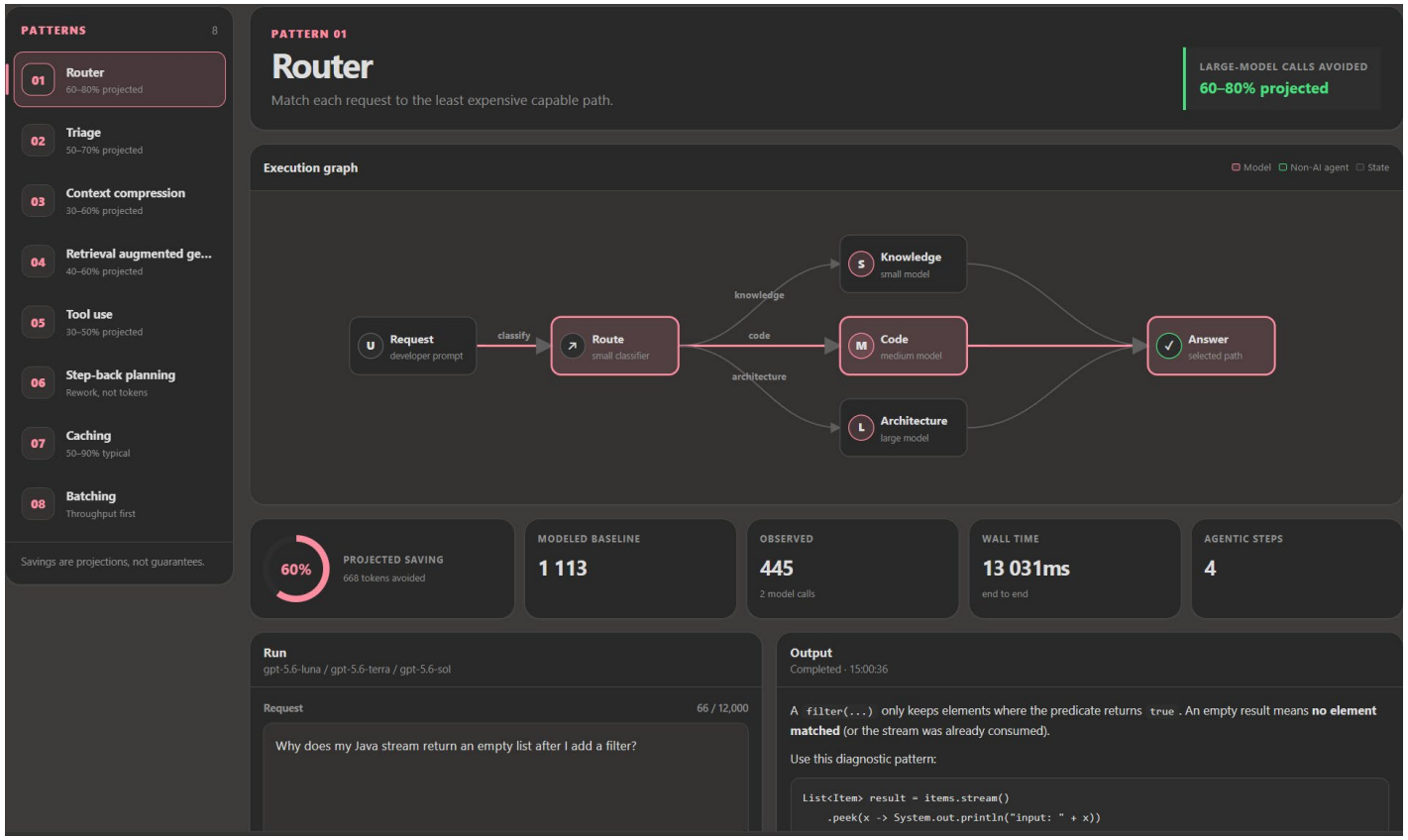




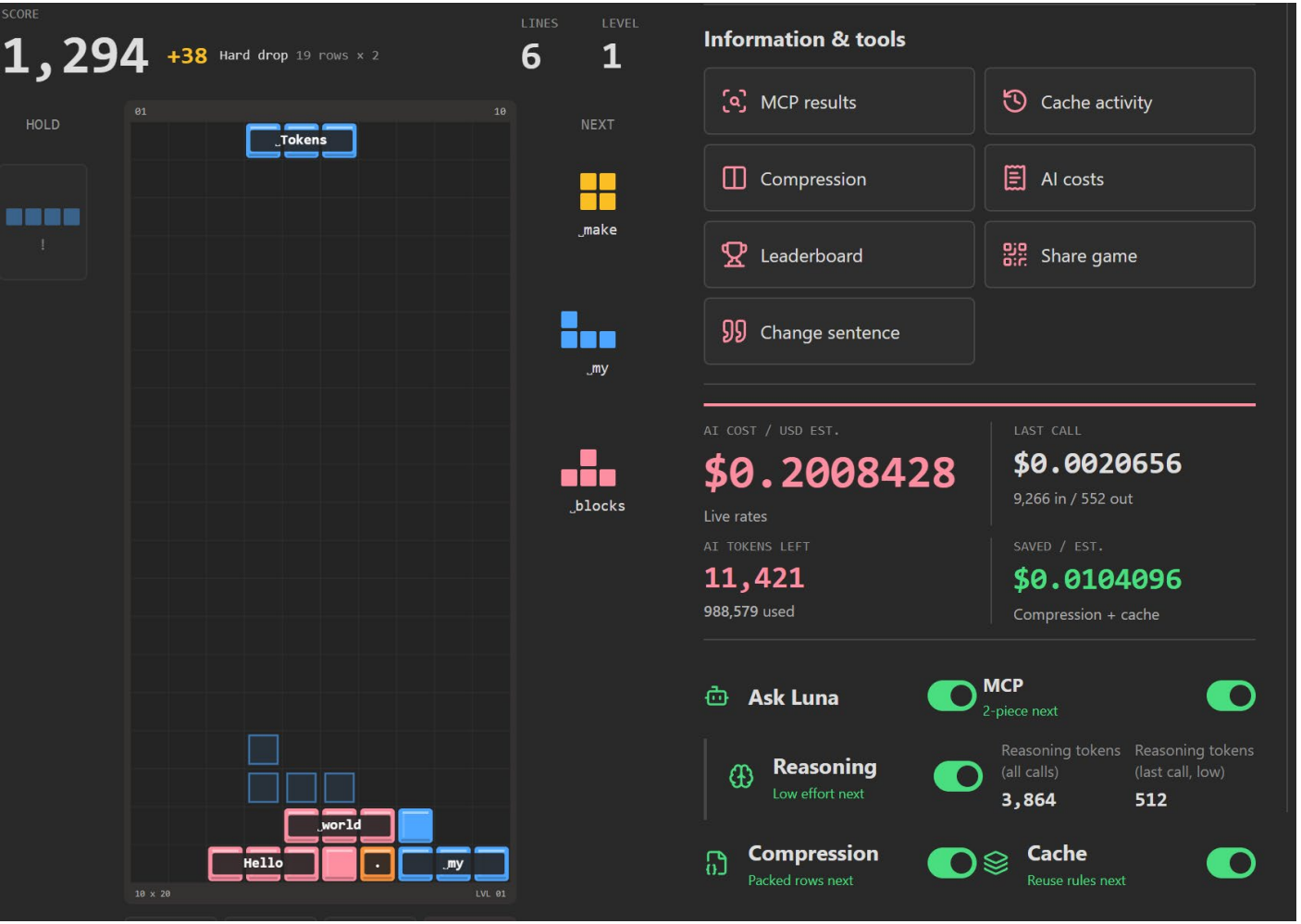




https://aka.ms/token-economics



https://github.com/roryp/token-design-patterns



https://github.com/roryp/tokenfall